ETHGLOBAL MUMBAI · PRODUCT WALKTHROUGH V3 · 5 NOV 2026

Voucher Rail Walkthrough

A crypto holder who wants to shop gets a gift voucher. A card holder who wants crypto buys that voucher on her best reward route and keeps 100% of her card rewards. A verification layer checks the voucher is live before the escrow releases the USDT. No rupees move between strangers.

The two people in this example

SELLER OF CRYPTO

Rohan, Bengaluru

Holds 120 USDT. Wants to buy clothes worth ₹10,000 at a Lifestyle store this weekend without cashing out through an exchange.

BUYER OF CRYPTO

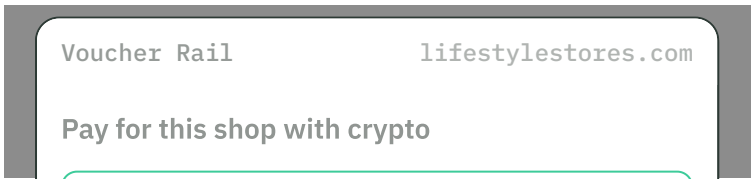
Priya, Pune

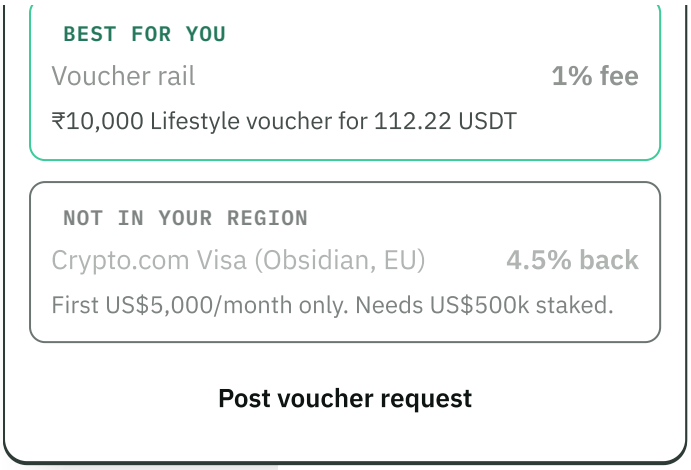
Holds HDFC Infinia (saved in her card portfolio). Wants about 110 USDT. Has not used her SmartBuy voucher bonus this month.

OUTCOME OF THE TRADE	ROHAN (SELLER)	PRIYA (BUYER)
Gives	112.22 USDT	₹10,000 on her card
Gets	₹10,000 Lifestyle vouchers	111.11 USDT + 1,650 reward points
Effective price per USDT	₹89.11	₹75.15 (travel value) · ₹85.55 (statement value)
Saving shown in the extension	**pays 1% fee, nothing else**	**16.5% (travel) · 4.95% (statement)**

Example rate ₹90 per USDT, set by the two users in the order. Card numbers from the Infinia terms and SmartBuy offer 15282 as checked on 4 Nov 2026: 5 points per ₹150, brand vouchers 5X, voucher bonus capped at 3,000 points a month, ₹1 per point on flights and hotels, up to ₹0.30 per point as statement credit. The live product re-checks these before every recommendation.

Step by step, screen by screen





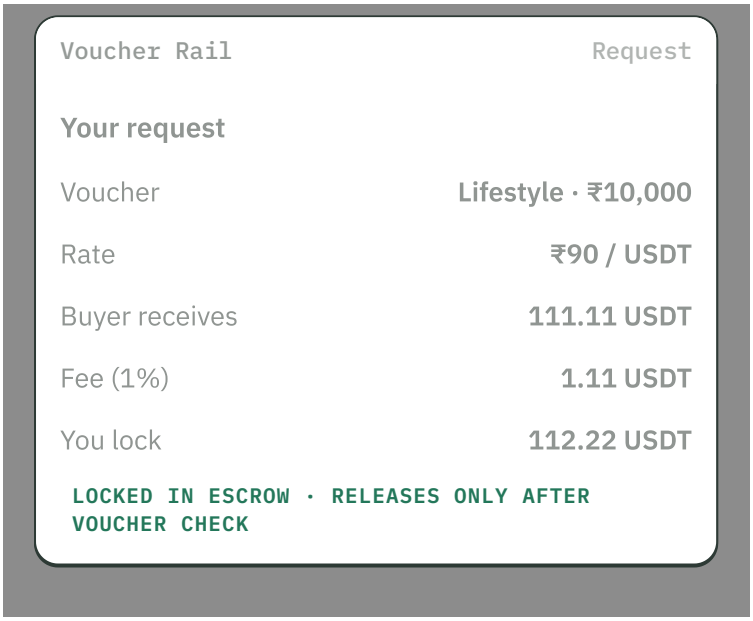
1 ROHAN The extension spots the merchant and offers the crypto options

Rohan opens Lifestyle's site to pick his clothes. The extension recognises the merchant and compares two ways to pay with crypto: our voucher rail, or a crypto prepaid card that pays rewards. The card is shown with its real limits. It isn't available in his region, so the voucher rail wins.

TECH

- Content script reads the domain only, never page data or card details
- Crypto card catalogue: reward rate, cap, staking needs and countries, with source and date checked
- Voucher catalogue maps the merchant to the voucher brands that work there

---



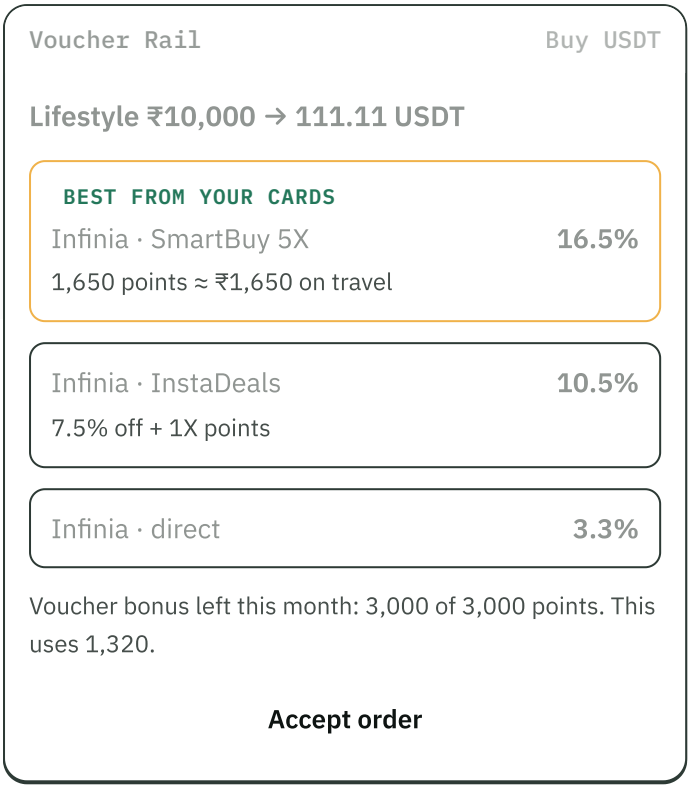
2 ROHAN He locks the USDT and the request goes live

He can take one of hundreds of suggested vouchers or type a custom one (any brand, any amount). His USDT moves into the escrow contract, so buyers know the crypto is really there.

TECH

- Escrow contract holds USDT against an order ID (brand, amount, rate, deadline)
- Request appears in the order book shown to buyers
- World ID check on first use: one real person per account

World ID check on first use. One real person per account



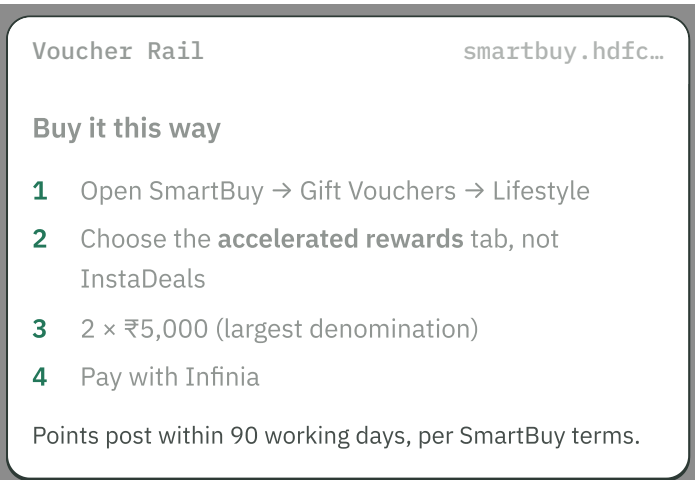
3 PRIYA Her extension ranks every way to buy that voucher

Priya wants USDT. She sees Rohan's request with the result already worked out from her saved cards: which card, which channel (SmartBuy, InstaDeals, other bank portals like Axis Grab Deals), and how much of her monthly cap she has left. She accepts.

TECH

- Card portfolio stored on her device only: issuer, variant, caps used. No card numbers
- Route engine: base rate × channel multiplier, minus caps, per-transaction rounding and exclusions
- Channel rates (SmartBuy, GyFTR, bank portals) refreshed from official pages with date checked

---

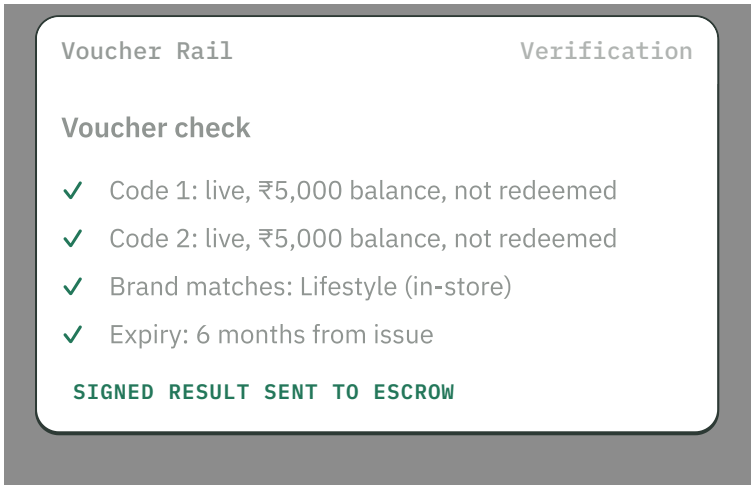


4 PRIYA She buys the vouchers on the recommended route

The extension walks her through the exact route so the 5X multiplier applies. She pays on SmartBuy as usual. The reward is between her and HDFC; we never touch it.

TECH

- Guided steps on the bank portal; she checks out herself
- Her local record of caps used updates after purchase



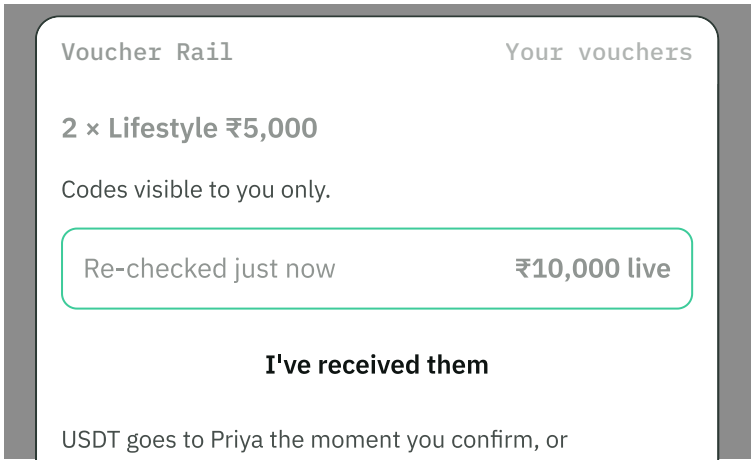
5 VERIFICATION LAYER The codes are checked live before Rohan sees them

Priya pastes the codes into the extension. They go encrypted to our verification layer, which checks balance and status on the issuer's balance-check page. For vouchers above ₹10,000, or brands like jewellers (Kalyan, Tanishq), it checks on the merchant's own portal. Only a pass moves the trade forward.

TECH

- Codes encrypted on Priya's device to the verifier key; deleted after the trade
- Verifier queries the issuer or merchant balance check, then signs: order ID, amount, status, time
- Escrow contract accepts only a signed pass. Stretch: a zkTLS proof of the balance page instead of trusting our signature

---



automatically 30 min after the re-check passes.

6 ROHAN Rohan gets the codes, the escrow pays Priya

The codes open only for Rohan. The verifier re-checks the balance at that moment, because Priya also has the codes. Then the contract sends 111.11 USDT to Priya and splits the 1.11 USDT fee. Rohan shops on Saturday.

TECH

- Codes re-encrypted to Rohan's key; second verifier check right before release
- Contract releases 111.11 USDT to Priya
- Fee split (proposal): 0.56 USDT to the dispute reserve, 0.55 USDT platform revenue

---

Voucher Rail Dispute

Voucher didn't work at the store?

1. Verifier pulls the voucher's history
2. Spent before your first use → you're refunded from the reserve